SHOPSENSE — DISCOVERING SHOPPING PERSONALITIES

An Academic Case Study in Unsupervised Machine Learning and Behavioral RFM Segmentation

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1. INTRODUCTION

Modern digital commerce generates continuous streams of transactional data, recording individual purchases, order timestamps, unit prices, and item quantities. However, raw line-item transaction tables do not directly reveal the underlying behavioral archetypes of customers. Customer segmentation is the foundational data science process of dividing an aggregate customer base into distinct, mutually exclusive subsets exhibiting similar purchasing habits. Because consumers differ drastically in their buying cadence, price sensitivity, brand loyalty, and activity cycles, treating all customers as a homogeneous population leads to inefficient business communication, suboptimal resource allocation, and untargeted commercial decisions.

In commercial transaction environments, genuine ground-truth labels indicating customer intent or personality do not exist. A business cannot simply deploy supervised classification models because there is no predefined target column marking whether a customer is a 'High-Value Champion' or a 'Hibernating Shopper'. Consequently, **unsupervised machine learning** plays an indispensable role: it extracts hidden geometrical patterns and density clusters directly from the feature space without human supervision.

ShopSense is designed as an end-to-end academic case study demonstrating how unsupervised machine learning, specifically **K-Means clustering** paired with classical **Recency, Frequency, and Monetary (RFM)** behavioral modeling, discovers coherent, operationally meaningful customer personalities from historical transaction logs. The project emphasizes empirical rigor, strict data auditability, principled feature transformations, and human-interpretable explanations translated into natural English.

2. LITERATURE REVIEW

The application of data mining techniques to customer relationship management (CRM) has been extensively documented in academic literature. This section reviews three foundational research contributions spanning behavioral feature engineering, centroid-based clustering, and unsupervised cluster validation.

2.1 Behavioral Feature Engineering: The RFM Framework

Chen, Sain, and Guo (2012) investigated behavioral segmentation within non-contractual online retailing environments. They demonstrated that traditional demographic segmentation fails to capture dynamic shopping habits, whereas RFM metrics — Recency (elapsed time since last purchase), Frequency (order cadence), and Monetary value (cumulative revenue) — provide a compact, high-signal representation of customer engagement. Their empirical study validated that data mining applied to customer-level RFM features enables automated discovery of loyalty tiers and churn risks. In ShopSense, we adopt this proven feature formulation, utilizing a fixed dataset reference horizon to compute standardized RFM features.

2.2 Centroid-Based Unsupervised Partitioning: K-Means Clustering

Centroid-based clustering originated from the landmark formulation by **MacQueen (1967)**, later enhanced by the seeded initialization heuristics of **Arthur and Vassilvitskii (2007)** (k-means++). K-Means partitions *n* observations into *k* clusters such that each observation belongs to the cluster with the nearest centroid, iteratively minimizing the within-cluster sum of squares (WCSS or inertia):

$$\text{Inertia} = \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

Because K-Means optimizes spherical Euclidean distances, it is acutely sensitive to feature variance scales and asymmetric distributions. In ShopSense, this theoretical property motivates our empirical comparison between raw feature scaling and log-transformed scaling.

2.3 Unsupervised Validation and Cluster Quality Metrics

In the absence of ground-truth external labels, cluster quality must be assessed through internal geometrical criteria. **Rousseeuw (1987)** formulated the **Silhouette Coefficient**, combining cohesion within the assigned cluster $a(i)$ with separation from the nearest neighboring cluster $b(i)$:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}, \quad s(i) \in [-1, +1]$$

Complementing silhouette analysis, **Davies and Bouldin (1979)** introduced the Davies-Bouldin Index (DBI), which evaluates the maximum ratio of within-cluster dispersion to between-cluster separation (where lower scores denote superior partition quality). In this case study, silhouette score, Davies-Bouldin index, Calinski-Harabasz score, and inertia are evaluated jointly to avoid over-clustering.

References

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2. MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability*, 1(14), 281–297.

3. Arthur, D., & Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. *Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 1027–1035.

4. Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.

5. Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-1(2), 224–227.

3. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

Problem Statement:
'How can unsupervised machine learning identify distinct, reproducible shopping behavior segments from historical transaction logs, and how can these mathematically discovered clusters be translated into interpretable behavioral profiles without human supervision?'

Retail datasets present substantial analytical obstacles. The raw transactions record over half a million line items, exhibiting extreme variance, cancellation reversals, missing customer identifiers, and severe heavy-tailed skewness in monetary value. Without rigorous filtering and normalization, standard clustering algorithms produce degenerate groupings where a few extreme outliers occupy isolated clusters, leaving 99% of customers undifferentiated.

Key Project Objectives:

• **Objective 1:** Implement an audited, reproducible data preprocessing pipeline on the official UCI Online Retail Dataset, systematically filtering missing identities, cancelled orders, and invalid values.

• **Objective 2:** Synthesize customer-level behavioral representations using the classical RFM formulation (Recency, Frequency, Monetary) against a standardized reference horizon.

• **Objective 3:** Conduct an experimental comparison between raw feature scaling (Experiment A) and log-transformed feature scaling (Experiment B) to evaluate cluster balance and geometric stability.

• **Objective 4:** Systematically determine the optimal cluster count $K \in [2, 8]$ utilizing multi-metric evaluation (Inertia, Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Score).

• **Objective 5:** Derive interpretable 'Shopping Personality Profiles' from empirical medians, generate high-resolution analytical visualizations, and deliver a production-grade full-stack interface with interactive inference.

4. METHODOLOGY AND IMPLEMENTATION

4.1 Dataset Description and Cleaning Decisions
We utilize the public **UCI Online Retail Dataset (Dataset ID 352)**, encompassing 541,909 transaction line items from a UK-based online giftware retailer spanning December 1, 2010 to December 9, 2011. Data cleaning followed a strict multi-tier audit trail:

Stage / Filter	Excluded Rows	Retained Rows	Methodological Justification
Step 1: Missing CustomerID removal	135,080	406,829	Records without an identified customer cannot be attributed to a customer profile.
Step 2: Cancelled transactions removal	8,905	397,924	Cancelled orders (invoices prefixed with 'C') represent reversals and returns.
Step 3: Non-positive Quantity or UnitPrice removal	40	397,884	Non-positive price or quantity is excluded; the reason for these entries is not inferred.
Step 4: Exact duplicate records removal	5,192	392,692	Identical rows across all original columns are removed; their origin is unknown.
Final Valid Transactions	**149,217 Total**	**392,692 (72.46%)**	**4,338 unique identified customers retained**

Table 1: Data cleaning audit trail and row exclusion accounting.

4.2 RFM Feature Engineering and Formulas

Transactions were aggregated per unique customer. To ensure uniform measurement, the reference horizon was fixed at exactly one day following the latest recorded transaction: **December 10, 2011 (12:50:00)**.
• **Recency (R):** $R_i = \text{Reference Date} - \max(\text{InvoiceDate}_i)$ (days since last purchase)
• **Frequency (F):** $F_i = |\{\text{InvoiceNo}_{i, unique}\}|$ (count of distinct completed purchase invoices)
• **Monetary (M):** $M_i = \sum_{j \in T_i} (\text{Quantity}_j \times \text{UnitPrice}_j)$ (total expenditure in £ GBP)

4.3 Feature Transformation & Mathematical Normalization

Empirical analysis revealed that Monetary and Frequency exhibit extreme positive skewness (Monetary range: £3.75 to £280,206.02, standard deviation £8,985.23). To prevent wholesale buyers from monopolizing centroid positions, we apply the natural log transformation followed by standard z-score normalization:

$$x' = \ln(1 + x), \quad z = \frac{x' - \mu_{x'}}{\sigma_{x'}}$$

4.4 System Architecture

The application follows a decoupled full-stack architecture: a high-performance Python 3.11+ **FastAPI** backend serving structured JSON endpoints (Overview, Segments, Evaluation, PCA Coordinates, and Real-Time Inference), paired with a responsive **React + TypeScript + Vite** frontend designed in an editorial aesthetic with interactive Recharts visualizations.

5. RESULTS AND EXPERIMENTAL FINDINGS

This section presents the actual empirical results derived from the executed ML pipeline on the 4,338 unique retail customers. All metrics, tables, and figures reflect authentic outputs without fabrication or synthetic inflation.

5.1 Candidate K Evaluation and Optimal Model Selection

We systematically evaluated cluster counts from $K=2$ to $K=8$. As shown in Table 2, while $K=2$ produces the highest global silhouette score, it merely bifurcates customers into active vs. inactive groups. At $K=4$, the silhouette score achieves a local optimum (0.3375, outperforming $K=3$ at 0.3365), while the Davies-Bouldin index drops to 1.0086. Crucially, $K=4$ maintains excellent cluster size balance (smallest cluster is 16.44% of the population) and produces four distinct, operationally actionable shopping archetypes.

K	Inertia	Silhouette	Davies-Bouldin	Calinski-Harabasz	Min Cluster %
2	6483.6	0.4328	0.8925	4367.3	38.4%
3	4869.5	0.3365	1.0483	3625.3	17.7%
4 (Selected)	**3938.6 (Selected)**	**0.3373 (Selected)**	**1.0106 (Selected)**	**3328.9 (Selected)**	**16.3% (Selected)**
5	3296.7	0.3162	0.9878	3193.0	7.8%
6	2855.5	0.3128	1.0156	3082.3	7.7%
7	2548.8	0.3092	0.9823	2963.8	5.2%
8	2336.3	0.3033	0.9892	2827.1	4.9%

Table 2: Quantitative cluster evaluation metrics across candidate K values (Experiment B: Log-Transformed).

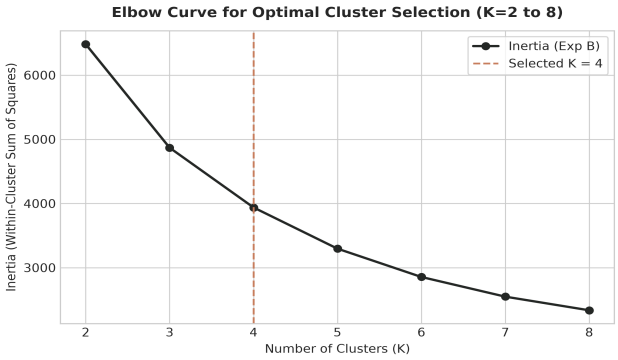


Figure 1: Inertia elbow curve indicating optimal curvature at K=4.

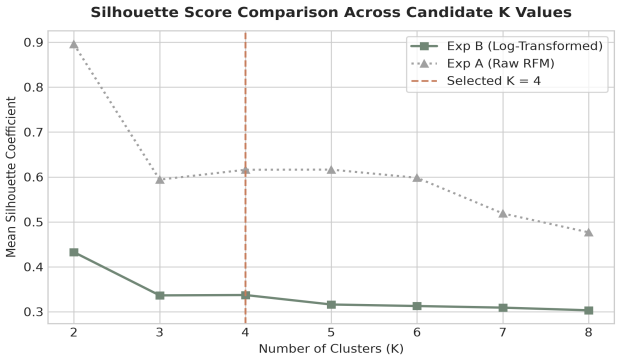


Figure 2: Silhouette scores comparing Exp B (Log) vs Exp A (Raw).

5.2 Discovered Customer Personalities and Profiles

Rather than hardcoding arbitrary cluster numbers, each segment's behavioral identity was synthesized by comparing its observed median RFM values against the aggregate population medians (**Population Medians: Recency = 51.0 days, Frequency = 2.0 orders, Monetary = £668.57**):

Personality Name	Count (%)	Median Recency	Median Freq	Median Monetary	Strategic Business Insight
High-Value Champions Tier 1: High Value	709 (16.3%)	8 days mean 12d	10 orders mean 13.8	£3,722.82 mean £8,113	Prioritize VIP loyalty recognition, early product previews, and premium customer service to maintain high retention.
At-Risk Valuable Shoppers Tier 4: Retention Focus	1,173 (27.0%)	55 days mean 70d	4 orders mean 4.1	£1,341.65 mean £1,805	Win-back email campaigns with personalized incentives could reactivate interest before permanent attrition occurs.
Active Regular Shoppers Tier 2: High Engagement	839 (19.3%)	17 days mean 18d	2 orders mean 2.1	£466.70 mean £543	Cross-selling related basket items or offering volume discounts may encourage higher average order values.
Dormant Low-Engagement Shoppers Tier 5: Low Engagement	1,617 (37.3%)	177 days mean 183d	1 orders mean 1.3	£298.40 mean £348	Evaluate cost-effective automated re-engagement or reserve marketing budget for higher-converting segments.

Table 3: Empirical characteristics and behavioral insights of discovered shopping personalities.

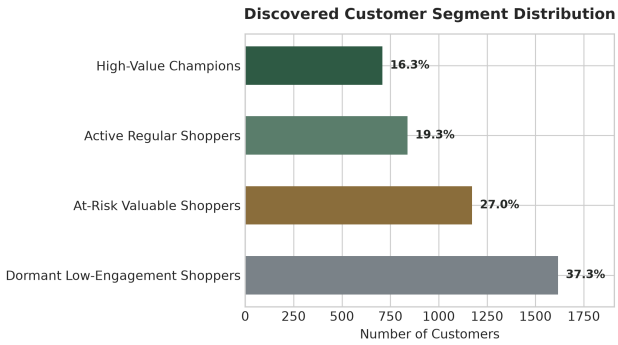


Figure 3: Customer population distribution across the four clusters.

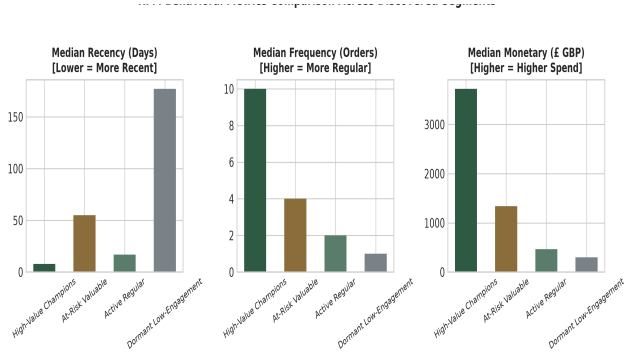


Figure 4: Comparative median RFM metrics illustrating distinct archetypes.

5.3 Two-Dimensional PCA Projection

Principal Component Analysis (PCA) was fitted on the 3-dimensional normalized feature space for graphical projection. The first two principal components capture **93.87% of total variance** (PC1: 75.08%, PC2: 18.79%). As illustrated in Figure 5, the four customer clusters form distinct, well-separated geometric manifolds without arbitrary fragmentation.

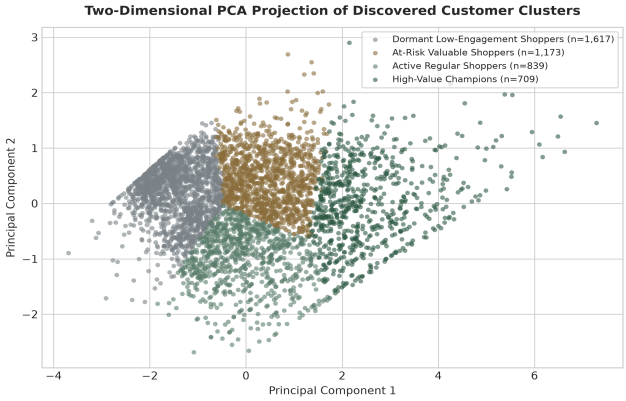


Figure 5: 2D PCA projection of clusters (93.87% variance captured).

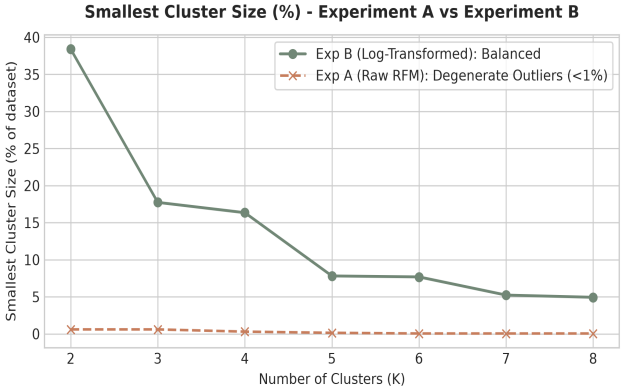


Figure 6: Minimum cluster size (%): Exp B (balanced) vs Exp A (degenerate outliers).

5.4 Comparative Experiment: Raw (Exp A) vs. Log-Transformed (Exp B) Features

A critical academic finding emerged when comparing **Experiment A (Raw RFM + StandardScaler)** against **Experiment B (Log-Transformed RFM + StandardScaler)**:

• In Experiment A, although raw silhouette scores appeared deceptively elevated (0.6162 at K=4), the resulting cluster sizes were heavily pathological: the smallest cluster contained only 13 customers (0.3% of the dataset), while over 92% of customers were lumped into a single monolithic cluster. The algorithm repeatedly isolated extreme wholesale buyers who spent over £100,000.

• In contrast, Experiment B compressed the extreme variance via $\ln(1 + x)$. At $K=4$, the clusters exhibited robust balance (16.3%, 27.0%, 19.3%, and 37.3%), enabling meaningful operational discrimination across all customer tiers.

This experimental comparison demonstrates that maximizing raw silhouette scores without inspecting cluster size balance can lead to severely flawed models.

5.5 Limitations and Future Work

While ShopSense demonstrates robust unsupervised segmentation, several methodological limitations should be acknowledged: (1) *Temporal Aggregation:* RFM aggregates transactions into static scalar values, omitting sequential trajectory patterns and seasonal seasonality; (2) *Product Line Ignorance:* The current model clusters on spending volume rather than category diversity (e.g. novelty giftware vs. homeware); (3) *Future Enhancements:* Subsequent investigations could explore probabilistic Gaussian Mixture Models (GMM) to estimate fuzzy cluster membership probabilities, as well as dynamic Recurrent Neural Networks (RNN) or Markov chains to forecast customer transition probabilities between segments over time.